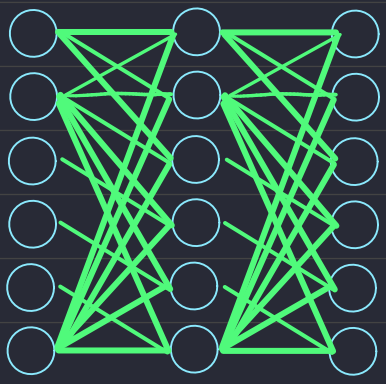
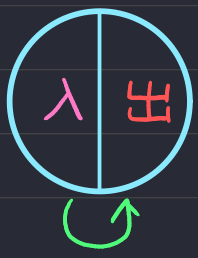
An Artificial Neuron - The Perceptron



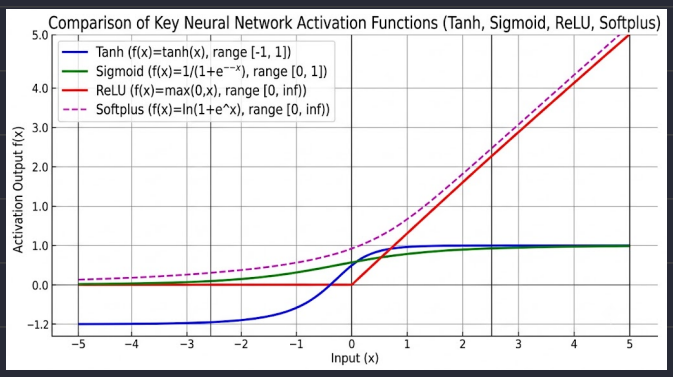
A Neuron

上層的 n 個神經輸出



1. 權重與位移： $f = w_1 \times in_1 + w_2 \times in_2 + w_n \times in_n + b$
2. 套用激勵函式 (Activation Function): e.g. $y = \mathrm{ReUL}(f)$

Activation Function



1. Tanh (Hyperbolic tangent)

$$f(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

2. Sigmoid / Logistic

$$f(x) = \sigma(x) = \frac{1}{1 + e^{-x}}$$

3. ReLu (Rectified Linear Units)

$$f(x) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$$

4. Softplus

$$f(x) = \ln(1 + e^x)$$

$$out_{o1} = \frac{1}{1 - e^{-x}}$$

$$\frac{\partial out_{o1}}{\partial x} = \frac{-(-e^{-x})}{(1 + e^{-x})^2} = \frac{e^{-}x}{(1 + e^{-x})^2} = \frac{(1 + e^{-x}) - 1}{(1 + e^{-x})^2} = \frac{1 + \frac{1}{1+e^{-x}}}{1 + e^{-x}} = (1 - f(x)) \times f(x)$$